Supplementary Material

This document contains the supplementary material to the work "Manifold Optimization on the Magnitude Torus for Fourier Phase Retrieval."

In Sec. A, we present the proofs of the theorems and corollaries appearing in the manuscript. In Sec. B, we provide additional numerical simulations of the `dualPGD` algorithm to demonstrate its applicability. Further remarks on our results are given in Sec. C.

A. PROOFS OF THEOREMS AND COROLLARIES

This section contains the detailed proofs of the theorems and corollaries appearing in the manuscript. These results are listed in the order in which they appear. Recall the following two assumptions:

Assumption 1. *Given $\boldsymbol{y} \in \mathbb{R}_{\geq 0}^{n\times 1}$, there exists an $\boldsymbol{x} \in \mathbb{A}_{\boldsymbol{y}} \cap \mathcal{B}$.*

Assumption 2. *The given $\boldsymbol{y} \in \mathbb{R}_{\geq 0}^{n\times 1}$ has no zero entries, i.e., $\boldsymbol{y} \in \mathbb{R}_{> 0}^{n\times 1}$.*

Theorem 1. *The gradient $\nabla\ell$ vanishes only at the global minima. The Hessian $H := \nabla^2\ell$ satisfies $H \cdot \nabla\ell = 2\nabla\ell$.*

Proof. Let us derive the gradient and the Hessian analytically, then the theorem follows immediately from their expressions. Let $(\cdot)^*$ denote complex conjugation. To simplify the notation, we denote $\boldsymbol{x}^{\odot n}$ by $\boldsymbol{x}^n$ for any vector $\boldsymbol{x}$ and exponent $n > 0$, and by $1/\boldsymbol{x}^{-n}$ for $n < 0$.

Throughout this proof, we restrict attention to vectors $\boldsymbol{x}$ such that $A\boldsymbol{x}$ has no zero entries, where ℓ is smooth.

For an arbitrary real vector $\boldsymbol{v}$ of compatible size, consider the directional derivative

$$\begin{aligned}
\frac{\partial |A\boldsymbol{x}|}{\partial \boldsymbol{x}}[\boldsymbol{v}] &= \frac{\partial}{\partial \boldsymbol{x}}\left(\sqrt{(A\boldsymbol{x})\odot(A^*\boldsymbol{x})}\right)[\boldsymbol{v}] \\
&= \frac{1}{2|A\boldsymbol{x}|}\big((A\boldsymbol{v})\odot(A^*\boldsymbol{x}) + (A\boldsymbol{x})\odot(A^*\boldsymbol{v})\big) \\
&= \frac{1}{|A\boldsymbol{x}|}\mathbf{Re}\big\{(A\boldsymbol{x})\odot(A^*\boldsymbol{v})\big\} \\
&= \frac{1}{|A\boldsymbol{x}|}\mathbf{Re}\big\{\operatorname{diag}(A\boldsymbol{x})(A^*\boldsymbol{v})\big\} \\
&= \frac{1}{|A\boldsymbol{x}|}\mathbf{Re}\big\{\operatorname{diag}(A\boldsymbol{x})A^*\big\}\,\boldsymbol{v}.
\end{aligned}$$

For the loss function defined as $\ell(\boldsymbol{x}) = \big\||A\boldsymbol{x}| - \boldsymbol{y}\big\|_2^2$, we have

$$\begin{aligned}
\frac{\partial \ell(\boldsymbol{x})}{\partial \boldsymbol{x}}[\boldsymbol{v}] &= 2\left(\frac{\partial |A\boldsymbol{x}|}{\partial \boldsymbol{x}}[\boldsymbol{v}]\right)^{\mathsf{T}}\big(|A\boldsymbol{x}| - \boldsymbol{y}\big) \\
&= 2\boldsymbol{v}^{\mathsf{T}}\mathbf{Re}\Big\{A^{\mathsf{H}}\operatorname{diag}(A\boldsymbol{x})\Big\}\frac{|A\boldsymbol{x}| - \boldsymbol{y}}{|A\boldsymbol{x}|} \\
&= 2\boldsymbol{v}^{\mathsf{T}}\mathbf{Re}\Big\{A^{\mathsf{H}}\Big(\operatorname{sgn}(A\boldsymbol{x})\odot\big(|A\boldsymbol{x}| - \boldsymbol{y}\big)\Big)\Big\}.
\end{aligned}$$

However, under Assumption 1, $\operatorname{sgn}(A\boldsymbol{x})\odot(|A\boldsymbol{x}| - \boldsymbol{y})$ is conjugate symmetric. Hence, by the properties of the Fourier transform, the expression inside $\mathbf{Re}\{\cdot\}$ is real-valued. Equating the expression above with $\boldsymbol{v}^{\mathsf{T}}\nabla\ell(\boldsymbol{x})$, we obtain the gradient of ℓ as

$$\nabla\ell(\boldsymbol{x}) = 2A^{\mathsf{H}}\big(\operatorname{sgn}(A\boldsymbol{x})\odot(|A\boldsymbol{x}| - \boldsymbol{y})\big). \tag{A1}$$

Note that the gradient is a real vector and that $A^{\mathsf{T}} = A$, we can also express it as

$$\nabla\ell(\boldsymbol{x}) = 2A\big(\operatorname{sgn}(A^*\boldsymbol{x})\odot(|A\boldsymbol{x}| - \boldsymbol{y})\big). \tag{A2}$$

The first statement of the theorem is equivalent to determining all vectors $\boldsymbol{x}$ that satisfy

$$\begin{aligned}
&\big\|\nabla\ell(\boldsymbol{x})\big\|_2 = 0 \\
\Leftrightarrow\ &\Big\|A^{\mathsf{H}}\big(\operatorname{sgn}(A\boldsymbol{x})\odot(|A\boldsymbol{x}| - \boldsymbol{y})\big)\Big\|_2 = 0 \\
\Leftrightarrow\ &\big\|\operatorname{sgn}(A\boldsymbol{x})\odot(|A\boldsymbol{x}| - \boldsymbol{y})\big\|_2 = 0 \\
\Leftrightarrow\ &\big\||A\boldsymbol{x}| - \boldsymbol{y}\big\|_2 = 0.
\end{aligned}$$

The second and third lines above are by the unitarity of A and $\operatorname{sgn}(A\boldsymbol{x})\odot$ (equivalent to the matrix $\operatorname{diag}(\operatorname{sgn}(A\boldsymbol{x}))$), respectively. Thus, the gradient vanishes only at global minima.

We further calculate the value of the Hessian of ℓ acting on an arbitrary real vector $\boldsymbol{v}$ to be the directional derivative

$$\begin{aligned}
H(\boldsymbol{x})\boldsymbol{v} &= \frac{\partial}{\partial \boldsymbol{x}}\big(\nabla\ell(\boldsymbol{x})\big)[\boldsymbol{v}] \\
&= 2A^{\mathsf{H}}\Bigg(\frac{A\boldsymbol{v}}{|A\boldsymbol{x}|}\odot\big(|A\boldsymbol{x}| - \boldsymbol{y}\big) \\
&\qquad - \frac{A\boldsymbol{x}}{|A\boldsymbol{x}|^2}\odot\big(|A\boldsymbol{x}| - \boldsymbol{y}\big)\odot\frac{\partial |A\boldsymbol{x}|}{\partial \boldsymbol{x}}[\boldsymbol{v}] \\
&\qquad + \operatorname{sgn}(A\boldsymbol{x})\odot\frac{\partial |A\boldsymbol{x}|}{\partial \boldsymbol{x}}[\boldsymbol{v}]\Bigg) \\
&= 2\boldsymbol{v} - A^{\mathsf{H}}\left(\boldsymbol{y}\odot\frac{A\boldsymbol{v}}{|A\boldsymbol{x}|}\right) \\
&\qquad + A^{\mathsf{H}}\left(\boldsymbol{y}\odot\frac{\operatorname{sgn}(A\boldsymbol{x})^2}{|A\boldsymbol{x}|}\odot(A^*\boldsymbol{v})\right).
\end{aligned}$$

Plugging in $\boldsymbol{v} = \nabla\ell(\boldsymbol{x})$, we have

$$\begin{aligned}
H(\boldsymbol{x})\boldsymbol{v} &= 2\boldsymbol{v} - 2A^{\mathsf{H}}\left(\boldsymbol{y}\odot\frac{A\boldsymbol{x}}{|A\boldsymbol{x}|^2}\odot(|A\boldsymbol{x}| - \boldsymbol{y})\right) \\
&\qquad + 2A^{\mathsf{H}}\left(\boldsymbol{y}\odot\frac{(A\boldsymbol{x})^2}{|A\boldsymbol{x}|^3}\odot\frac{A^*\boldsymbol{x}}{|A\boldsymbol{x}|}\odot(|A\boldsymbol{x}| - \boldsymbol{y})\right) \\
&= 2\boldsymbol{v}.
\end{aligned}$$

Hence, the theorem is proven. ∎

It is worthwhile to see that the function ℓ is also 2-PŁ (Polyak-Łojasiewicz):

$$\frac{1}{2}\big\|\nabla\ell(\boldsymbol{x})\big\|_2^2 = 2\cdot\big(\ell(\boldsymbol{x})-0\big)\,. \tag{A3}$$

The convergence of the gradient descent algorithm for minimizing ℓ over $\mathbb{R}^{n\times 1}$ is immediately obtained.

Theorem 2. *The set $\mathbb{A}_{\boldsymbol{y}}$ is the union of $2^{|\mathcal{N}_p|}$ disconnected tori each diffeomorphic to the d-torus $\mathbb{T}^d$, where*

$$\log_2\big|\mathcal{N}_p\big| = \#\{k\in[D] \mid N_k \text{ is even}\} \tag{7}$$

and

$$d:=|\mathcal{N}_c| = \frac{n-|\mathcal{N}_p|}{2}. \tag{8}$$

Proof. First, we determine the sets $\mathcal{N}_p$ and $\mathcal{N}_c$. We then establish the mutual orthogonality of $\{\boldsymbol{u}_k,\boldsymbol{v}_k\}_{k\in\mathcal{N}_p\cup\mathcal{N}_c}$. Finally, we will discuss the geometry of the intersections.

(a) The sets $\mathcal{N}_p$ and $\mathcal{N}_c$: For each $k\in[D]$, the Fourier matrix F_{N_k} contains two real row vectors when N_k is even, namely

$$[1,1,1,\dots] \quad \text{and} \quad [1,-1,1,\dots],$$

and contains only one real row vector when N_k is odd, namely

$$[1,1,1,\dots].$$

By definition, $\mathcal{N}_p$ consists of the indices in $[n]$ for which the corresponding row vector $\boldsymbol{a}_k$ of A is real. Since the real row vectors of A arise from Kronecker products of the real row vectors of the matrices F_{N_k}, we obtain

$$\big|\mathcal{N}_p\big| = 2^{\#\{\,k\in[D]\,\mid\, N_k \text{ is even}\,\}}.$$

The remaining indices in $[n]\backslash\mathcal{N}_p$ correspond to complex row vectors $\boldsymbol{a}_k$. Note that both $\boldsymbol{a}_k$ and $\boldsymbol{a}_k^*$ produce the same rank-2 matrix $A_k = \mathbf{Re}\{\boldsymbol{a}_k\boldsymbol{a}_k^{\mathsf{H}}\}$. Hence, for indices $j,k\in[n]\backslash\mathcal{N}_p$ with $j\neq k$ that satisfy $\boldsymbol{a}_j^*=\boldsymbol{a}_k$, we pick only one of $\{j,k\}$ and place it into $\mathcal{N}_c$. Evidently,

$$|\mathcal{N}_c| = \frac{n-|\mathcal{N}_p|}{2}.$$

(b) Orthogonality of $\boldsymbol{u}_k$'s and $\boldsymbol{v}_k$'s: By definition, we have

$$\boldsymbol{u}_k := \frac{1}{2}(\boldsymbol{a}_k+\boldsymbol{a}_k^*),$$
$$\boldsymbol{v}_k := \frac{1}{2\mathrm{i}}(\boldsymbol{a}_k-\boldsymbol{a}_k^*)$$

for all $k\in\mathcal{N}_p\cup\mathcal{N}_c$. We should first discuss the products of two $\boldsymbol{a}_k$'s.

By definition, $\boldsymbol{a}_k^{\mathsf{H}}$ is the kth row of A, the normalized DFT matrix. Since A is unitary, we have

$$\boldsymbol{a}_k^{\mathsf{H}}\boldsymbol{a}_j = \delta_{kj}$$

for all $k,j\in[n]$, where δ_{kj} denotes the Kronecker delta. Consider $k,j\in\mathcal{N}_p\cup\mathcal{N}_c$ with $j\neq k$, we also have the equality

$$\boldsymbol{a}_k^{\mathsf{T}}\boldsymbol{a}_j = (\boldsymbol{a}_k^*)^{\mathsf{H}}\boldsymbol{a}_j = 0.$$

Furthermore, since $\boldsymbol{a}_k = \frac{1}{\sqrt{n}}\bigotimes_{j\in[D]}\boldsymbol{f}_j^{(k)}$ for some collection of *Fourier vectors* $\{\boldsymbol{f}_j^{(k)}\}_{k\in[D]}$ with $\boldsymbol{f}_j^{(k)}$ being of the form $[1,\omega_j,\dots,\omega_j^{N_k-1}]^{\mathsf{T}}$ with $\omega_j^{N_j}=1$,

$$\begin{aligned}
\boldsymbol{a}_k^{\mathsf{T}}\boldsymbol{a}_k &= \frac{1}{n}\prod_{j\in[D]}\boldsymbol{f}_j^{(k)\mathsf{T}}\boldsymbol{f}_j^{(k)}\\
&= \frac{1}{n}\prod_{j\in[D]}\Big(1+\omega_j^2+\cdots+\omega_j^{2(N_j-1)}\Big)\\
&= \begin{cases}1, & \omega_j^2=1 \text{ for all } j\in[D];\\ 0, & \text{otherwise.}\end{cases}
\end{aligned}$$

Or equivalently,

$$\boldsymbol{a}_k^{\mathsf{T}}\boldsymbol{a}_k = \begin{cases}1, & k\in\mathcal{N}_p;\\ 0, & \text{otherwise.}\end{cases}$$

Combining the results above, we see that for all $k,j\in\mathcal{N}_p\cup\mathcal{N}_c$, we have

$$\begin{aligned}
\boldsymbol{u}_k^{\mathsf{T}}\boldsymbol{u}_j &= \frac{1}{4}\Big(\boldsymbol{a}_k^{\mathsf{T}}\boldsymbol{a}_j+\boldsymbol{a}_k^{\mathsf{T}}\boldsymbol{a}_j^*+\boldsymbol{a}_k^{\mathsf{H}}\boldsymbol{a}_j+\boldsymbol{a}_k^{\mathsf{H}}\boldsymbol{a}_j^*\Big)\\
&= \frac{1}{4}\Big(\boldsymbol{a}_k^{\mathsf{T}}\boldsymbol{a}_j+\boldsymbol{a}_k^{\mathsf{H}}\boldsymbol{a}_j^*+2\delta_{kj}\Big)\\
&= \frac{\delta_{kj}}{2}\Big(1+\mathbf{Re}\Big\{\boldsymbol{a}_k^{\mathsf{T}}\boldsymbol{a}_k\Big\}\Big)\\
&= \begin{cases}1, & k=j\in\mathcal{N}_p;\\ \frac{1}{2}, & k=j\in\mathcal{N}_c;\\ 0, & k\neq j,\end{cases}\\
\boldsymbol{v}_k^{\mathsf{T}}\boldsymbol{v}_j &= -\frac{1}{4}\Big(\boldsymbol{a}_k^{\mathsf{T}}\boldsymbol{a}_j-\boldsymbol{a}_k^{\mathsf{T}}\boldsymbol{a}_j^*-\boldsymbol{a}_k^{\mathsf{H}}\boldsymbol{a}_j+\boldsymbol{a}_k^{\mathsf{H}}\boldsymbol{a}_j^*\Big)\\
&= \frac{\delta_{kj}}{2}\Big(1-\mathbf{Re}\Big\{\boldsymbol{a}_k^{\mathsf{T}}\boldsymbol{a}_k\Big\}\Big)\\
&= \begin{cases}\frac{1}{2}, & k=j\in\mathcal{N}_c;\\ 0, & \text{otherwise,}\end{cases}\\
\boldsymbol{u}_k^{\mathsf{T}}\boldsymbol{v}_j &= \frac{1}{4\mathrm{i}}\Big(\boldsymbol{a}_k^{\mathsf{T}}\boldsymbol{a}_j-\boldsymbol{a}_k^{\mathsf{T}}\boldsymbol{a}_j^*+\boldsymbol{a}_k^{\mathsf{H}}\boldsymbol{a}_j-\boldsymbol{a}_k^{\mathsf{H}}\boldsymbol{a}_j^*\Big)\\
&= \frac{1}{4\mathrm{i}}\Big(\boldsymbol{a}_k^{\mathsf{T}}\boldsymbol{a}_j-\boldsymbol{a}_k^{\mathsf{H}}\boldsymbol{a}_j^*\Big)\\
&= \frac{1}{2}\mathbf{Im}\Big\{\boldsymbol{a}_k^{\mathsf{T}}\boldsymbol{a}_j\Big\}\\
&= 0.
\end{aligned}$$

Thus, the mutual orthogonality is established.

(c) Torus geometry: By the mutual orthogonality of the vectors $\boldsymbol{u}_k$ and $\boldsymbol{v}_k$, these vectors form a basis in $\mathbb{R}^{n\times 1}$. Each constraint $\left|\boldsymbol{a}_k^{\mathsf{H}}\boldsymbol{x}\right| = y_k$ thus only restricts the components of $\boldsymbol{x}$ in $\operatorname{span}\{\boldsymbol{u}_k, \boldsymbol{v}_k\}$.

For each $k \in \mathcal{N}_p$, the solution set of $\left|\boldsymbol{a}_k^{\mathsf{H}}\boldsymbol{x}\right| = y_k$ consists of two parallel hyperplanes, namely $\boldsymbol{u}_k^{\mathsf{T}}\boldsymbol{x} = y_k$ and $\boldsymbol{u}_k^{\mathsf{T}}\boldsymbol{x} = -y_k$, the number of connected components of $\mathbb{A}_{\boldsymbol{y}}$ is $2^{\left|\mathcal{N}_p\right|}$ under the nondegeneracy assumption on $\boldsymbol{y}$ (Assumption 2). For $k \in \mathcal{N}_c$, on the other hand, the constraint $\left|\boldsymbol{a}_k^{\mathsf{H}}\boldsymbol{x}\right| = y_k$ removes one degree of freedom of $\boldsymbol{x}$ within $\operatorname{span}\{\boldsymbol{u}_k, \boldsymbol{v}_k\}$ to trace out a circle. Thus, the intersection forms a torus with

$$n - \left|\mathcal{N}_p\right| - \left|\mathcal{N}_c\right| = d$$

degrees of freedom, i.e., a d-torus. ∎

Corollary 1. *Trivial associates of a signal $\boldsymbol{x}$ obtained by odd translations along dimensions of even lengths, or by global sign-flip, lie on different tori from the one containing $\boldsymbol{x}$. All other trivial associates lie on the same torus.*

Proof. There are three types of trivial associates: translations, (point) reflection, and global sign-flip. The latter trivially lies on a different torus from the original signal.

Let P be a permutation matrix such that $\boldsymbol{x}' = P\boldsymbol{x}$ is a trivial associate of $\boldsymbol{x}$. For $\boldsymbol{x}'$ to lie on a different torus from $\boldsymbol{x}$, we need

$$\boldsymbol{a}_k^{\mathsf{H}}\boldsymbol{x} = -\boldsymbol{a}_k^{\mathsf{H}}\boldsymbol{x}' = -\boldsymbol{a}_k^{\mathsf{H}}P\boldsymbol{x} \Leftrightarrow \boldsymbol{a}_k = -P^{\mathsf{T}}\boldsymbol{a}_k$$

for some $k \in \mathcal{N}_p$. Recall that $\boldsymbol{a}_k = \frac{1}{\sqrt{N}} \bigotimes_{j\in[D]} \boldsymbol{f}_j^{(k)}$. The sign flip $P^{\mathsf{T}}\boldsymbol{a}_k = -\boldsymbol{a}_k$ occurs if and only if $\boldsymbol{f}_j^{(k)}$ takes the alternating form $[1, -1, 1, \ldots]^{\mathsf{T}}$ (which requires N_j to be even) and P acts as an odd translation in that direction. For even translations or reflection ($\boldsymbol{x}[n] \mapsto \boldsymbol{x}[-n]$), no such sign change is induced. ∎

Theorem 3. *Any point $\boldsymbol{x} \in \mathbb{A}_{\boldsymbol{y}}$ can be uniquely parameterized by $\boldsymbol{s} = [s_1, s_2, \ldots] \in \{1, -1\}^{\left|\mathcal{N}_p\right|}$ and $\boldsymbol{t} = [t_1, t_2, \ldots]^{\mathsf{T}} \in [0, 2\pi)^d = \mathbb{T}^d$ via*

$$\boldsymbol{x}(\boldsymbol{s}, \boldsymbol{t}) = \sum_{j\in\mathcal{N}_p} s_j y_j \boldsymbol{u}_j + \sum_{k\in\mathcal{N}_c} 2y_k \left(\cos t_k \boldsymbol{u}_k + \sin t_k \boldsymbol{v}_k\right). \tag{9}$$

Proof. One can check that the above parameterization indeed satisfies all constraints $\left|\boldsymbol{a}_k^{\mathsf{H}}\boldsymbol{x}\right| = y_k$, $k \in \mathcal{N}_p \cup \mathcal{N}_c$. Here, however, we provide a more constructive proof by building on part (c) in the proof of Theorem 2.

Since $\{\boldsymbol{u}_k, \boldsymbol{v}_k\}_{k\in\mathcal{N}_p\cup\mathcal{N}_c}$ is a basis of $\mathbb{R}^{n\times 1}$, any vector can be expressed as a linear combination of the basis vectors. Assume

$$\boldsymbol{x} = \sum_{j\in\mathcal{N}_p} a_j \boldsymbol{u}_j + \sum_{k\in\mathcal{N}_c} \left(b_k \boldsymbol{u}_k + c_k \boldsymbol{v}_k\right),$$

we have

$$a_j = \frac{\boldsymbol{u}_j^{\mathsf{T}}\boldsymbol{x}}{\boldsymbol{u}_j^{\mathsf{T}}\boldsymbol{u}_j} = \frac{\boldsymbol{a}_j^{\mathsf{H}}\boldsymbol{x}}{1} \in \{y_j, -y_j\},$$

which we denote the sign by s_j. For b_k and c_k ($k \in \mathcal{N}_c$), it is required that

$$\begin{aligned} y_k^2 &= \boldsymbol{x}^{\mathsf{T}} \left(\boldsymbol{u}_k\boldsymbol{u}_k^{\mathsf{T}} + \boldsymbol{v}_k\boldsymbol{v}_k^{\mathsf{T}}\right) \boldsymbol{x} \\ &= \frac{1}{4}(b_k^2 + c_k^2). \end{aligned}$$

A natural solution would be to set $b_k = 2y_k \cos t_k$ and $c_k = 2y_k \sin t_k$ for some $t_k \in [0, 2\pi)$. Thus, we arrive at the torus parameterization. ∎

Note that $\boldsymbol{t}$ is exactly the phase that we wish to retrieve in the Fourier phase retrieval problem.

Corollary 2. *Let $\boldsymbol{x} = \boldsymbol{x}(\boldsymbol{t})$ be the parameterization specified in Theorem 3,* $\operatorname{sgn}(\boldsymbol{a}_k^{\mathsf{H}}\boldsymbol{x}) = \mathrm{e}^{-\mathrm{i}t_k}$ *for all $k \in \mathcal{N}_c$.*

Proof. For $k \in \mathcal{N}_c$, we have $\boldsymbol{a}_k = \boldsymbol{u}_k + \mathrm{i}\boldsymbol{v}_k$. Then, by the orthogonality of $\{\boldsymbol{u}_k, \boldsymbol{v}_k\}_{k\in\mathcal{N}_p\cup\mathcal{N}_c}$,

$$\begin{aligned} \boldsymbol{a}_k^{\mathsf{H}}\boldsymbol{x} &= 2y_k \left(\boldsymbol{u}_k - \mathrm{i}\boldsymbol{v}_k\right)^{\mathsf{T}} \left(\cos t_k \boldsymbol{u}_k + \sin t_k \boldsymbol{v}_k\right) \\ &= 2y_k \left(\cos t_k \boldsymbol{u}_k^{\mathsf{T}}\boldsymbol{u}_k - \mathrm{i} \sin t_k \boldsymbol{v}_k^{\mathsf{T}}\boldsymbol{v}_k\right) \\ &= y_k \left(\cos t_k - \mathrm{i}\sin t_k\right) \\ &= y_k \mathrm{e}^{-\mathrm{i}t_k}. \end{aligned}$$

Thus it is shown. ∎

Corollary 3. *Let $P_{\mathcal{A}}$ be a projection onto $\mathbb{A}_{\boldsymbol{y}}$ defined by*

$$P_{\mathcal{A}}(\boldsymbol{x}') := \operatorname*{arg\,min}_{\boldsymbol{x}\in\mathbb{A}_{\boldsymbol{y}}} \left\|\boldsymbol{x} - \boldsymbol{x}'\right\|_2^2. \tag{10}$$

Then, for points where the projection is unique,

$$P_{\mathcal{A}}(\boldsymbol{x}') = A^{-1} \left(\boldsymbol{y} \odot \operatorname{sgn}\left(A\boldsymbol{x}'\right)\right). \tag{11}$$

Proof. Without loss of generality, suppose

$$P_{\mathcal{A}}(\boldsymbol{x}') = \sum_{j\in\mathcal{N}_p} s_j y_j \boldsymbol{u}_j + \sum_{k\in\mathcal{N}_c} 2y_k \left(\cos t_k \boldsymbol{u}_k + \sin t_k \boldsymbol{v}_k\right)$$

and

$$\boldsymbol{x}' = \sum_{j\in\mathcal{N}_p} y_j' \boldsymbol{u}_j + \sum_{k\in\mathcal{N}_c} 2y_k' \left(\cos t_k' \boldsymbol{u}_k + \sin t_k' \boldsymbol{v}_k\right)$$

with $s_j \in \{1, -1\}$, $y_j' \in \mathbb{R}$, $y_k' > 0$, and $t_k \in \mathbb{R}$. Then by the orthogonality of $\boldsymbol{u}_k$'s and $\boldsymbol{v}_k$'s, the minimization problem amounts to solving

$$s_j = \operatorname*{arg\,min}_{s\in\{1,-1\}} \left|s y_j - y_j'\right|^2 = \operatorname{sgn}(y_j')$$

and

$$\begin{aligned} t_k &= \arg\min_{t\in\mathbb{R}} \Big\|(y_k\cos t - y_k'\cos t_k')\boldsymbol{u}_k \\ &\qquad + (y_k\sin t - y_k'\sin t_k')\boldsymbol{v}_k\Big\|^2 \\ &= \arg\min_{t\in\mathbb{R}} (y_k\cos t - y_k'\cos t_k')^2 \\ &\qquad + (y_k\sin t - y_k'\sin t_k')^2 \\ &= t_k'. \end{aligned}$$

Therefore, we see that $P_{\mathcal{A}}$ preserves the sign in the Fourier domain:

$$\begin{aligned} &\operatorname{sgn}\left(A\cdot P_{\mathcal{A}}(\boldsymbol{x}')\right) = \operatorname{sgn}(A\cdot\boldsymbol{x}') \\ &\Rightarrow A\cdot P_{\mathcal{A}}(\boldsymbol{x}') = \boldsymbol{y}\odot\operatorname{sgn}(A\boldsymbol{x}') \\ &\Rightarrow P_{\mathcal{A}}(\boldsymbol{x}') = A^{-1}\left(\boldsymbol{y}\odot\operatorname{sgn}(A\boldsymbol{x}')\right). \end{aligned}$$

The statement is thus proven. ∎

Theorem 4. *The tangent space and normal space of $\mathbb{A}_{\boldsymbol{y}}$ at $\boldsymbol{x}$ are characterized respectively by*

$$T_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}} = \ker\Big(\mathbf{Re}\big\{\operatorname{diag}(A\boldsymbol{x})A^*\big\}\Big) \tag{12}$$

and

$$N_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}} = \ker\Big(\mathbf{Im}\big\{\operatorname{diag}(A\boldsymbol{x})A^*\big\}\Big). \tag{13}$$

Note that there are multiple ways to show the results, here we present the ones as hinted in the manuscript.

Proof. We prove the two characterizations separately.

(a) Tangent space: For the characterization of the tangent space, this is simply an application of the regular value theorem: the magnitude torus is defined by $f(\boldsymbol{x}) = \mathbf{0}$, where $f:\mathbb{R}^{n\times 1}\to\mathbb{R}^{n\times 1}, \boldsymbol{x}\mapsto|A\boldsymbol{x}|^2-\boldsymbol{y}^2$. Let $\boldsymbol{x}$ be a regular point of f, then

$$T_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}} := \ker\left(\frac{\partial f}{\partial\boldsymbol{x}}\right).$$

The pushforward map $\partial f/\partial\boldsymbol{x}$ can be written in a matrix form: since for any $\boldsymbol{v}\in\mathbb{R}^{n\times 1}$,

$$\begin{aligned} \frac{\partial f}{\partial\boldsymbol{x}}[\boldsymbol{v}] &= \frac{\partial}{\partial\boldsymbol{x}}\left((A\boldsymbol{x})\odot(A\boldsymbol{x})^*\right)[\boldsymbol{v}] \\ &= (A\boldsymbol{x})\odot(A\boldsymbol{v})^* + (A\boldsymbol{v})\odot(A\boldsymbol{x})^* \\ &= 2\mathbf{Re}\big\{(A\boldsymbol{x})\odot(A^*\boldsymbol{v})\big\} \\ &= 2\mathbf{Re}\big\{\operatorname{diag}(A\boldsymbol{x})A^*v\big\} \\ &= 2\mathbf{Re}\big\{\operatorname{diag}(A\boldsymbol{x})A^*\big\}\,v. \end{aligned}$$

The regularity of $\boldsymbol{x}$ is guaranteed by Assumption 2. The first equation is thus shown. ∎

(b) Normal space: For the characterization of the normal space, we show that the two sets are equivalent. First, the inclusion ($\subseteq$): Note that by definition

$$N_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}} := \left(T_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}}\right)^\perp,$$

where $(\cdot)^\perp$ is the orthogonal complement over $\mathbb{R}^{n\times 1}$. Since

$$\begin{aligned} T_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}} &= \ker\Big(\mathbf{Re}\big\{\operatorname{diag}(A\boldsymbol{x})A^*\big\}\Big) \\ &= \left(\operatorname{Col}\left(\mathbf{Re}\big\{\operatorname{diag}(A\boldsymbol{x})A^*\big\}^\mathsf{T}\right)\right)^\perp \\ &= \left(\operatorname{Col}\left(\mathbf{Re}\big\{A^*\operatorname{diag}(A\boldsymbol{x})\big\}\right)\right)^\perp, \end{aligned}$$

where $\operatorname{Col}(\cdot)$ is the column space, we obtain

$$\begin{aligned} N_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}} &= \operatorname{Col}\Big(\mathbf{Re}\big\{A^*\operatorname{diag}(A\boldsymbol{x})\big\}\Big) \\ &= \operatorname{span}\Big\{\mathbf{Re}\Big\{\boldsymbol{a}_k\cdot y_k\mathrm{e}^{-\mathrm{i}t_k}\Big\}\Big\}_{k\in[n]} \\ &= \operatorname{span}\left\{\cos t_k\boldsymbol{u}_k + \sin t_k\boldsymbol{v}_k\right\}_{k\in[n]} \\ &= \operatorname{span}\left\{\cos t_k\boldsymbol{u}_k + \sin t_k\boldsymbol{v}_k\right\}_{k\in\mathcal{N}_p\cup\mathcal{N}_c}. \end{aligned}$$

Note the slight abuse of notation: $\mathrm{e}^{\mathrm{i}t_k} := s_k$ for $k\in\mathcal{N}_p$. This is equivalent to stating

$$N_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}} = \operatorname{span}\left\{\frac{\partial\boldsymbol{x}}{\partial y_k}\right\}_{k\in\mathcal{N}_p\cup\mathcal{N}_c}.$$

Next, for any $v\in\mathbb{R}^{n\times 1}$ and $j\in[n]$, the jth row of

$$\mathbf{Im}\big\{\operatorname{diag}(A\boldsymbol{x})A^*\big\}\,\boldsymbol{v}$$

is

$$\mathbf{Im}\Big\{(\boldsymbol{a}_j^\mathsf{H}\boldsymbol{x})\boldsymbol{a}_j^\mathsf{T}\Big\}\,\boldsymbol{v}.$$

Then, let $\boldsymbol{v}$ be the kth basis vector of $N_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}}$ ($k\in\mathcal{N}_p\cup\mathcal{N}_c$), i.e., $\boldsymbol{v} = \cos t_k\boldsymbol{u}_k + \sin t_k\boldsymbol{v}_k$, we have

$$\begin{aligned} &\mathbf{Im}\Big\{(\boldsymbol{a}_j^\mathsf{H}\boldsymbol{x})\boldsymbol{a}_j^\mathsf{T}\Big\}(\cos t_k\boldsymbol{u}_k + \sin t_k\boldsymbol{v}_k) \\ &= y_j\mathbf{Im}\Big\{\mathrm{e}^{-\mathrm{i}t_j}\boldsymbol{a}_j^\mathsf{T}(\cos t_k\boldsymbol{u}_k + \sin t_k\boldsymbol{v}_k)\Big\} \\ &= y_j\mathbf{Im}\Big\{\mathrm{e}^{-\mathrm{i}t_j}(\boldsymbol{u}_j + \mathrm{i}\boldsymbol{v}_j)^\mathsf{T}(\cos t_k\boldsymbol{u}_k + \sin t_k\boldsymbol{v}_k)\Big\} \\ &= y_k\delta_{jk}\mathbf{Im}\Big\{\mathrm{e}^{-\mathrm{i}t_k}\left(\cos t_k\boldsymbol{u}_k^\mathsf{T}\boldsymbol{u}_k + \mathrm{i}\sin t_k\boldsymbol{v}_k^\mathsf{T}\boldsymbol{v}_k\right)\Big\} \\ &= \frac{y_k\delta_{jk}}{2}\mathbf{Im}\{1\} = 0 \end{aligned}$$

by the orthogonality between the $\boldsymbol{u}_k$'s and $\boldsymbol{v}_k$'s. Therefore, by linearity, the inclusion

$$N_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}} \subseteq \ker\Big(\mathbf{Im}\big\{\operatorname{diag}(A\boldsymbol{x})A^*\big\}\Big)$$

is established.

For the reverse inclusion ($\supseteq$): Consider any vector $\boldsymbol{v} \in \ker(\mathbf{Im}\{\operatorname{diag}(A\boldsymbol{x})A^*\})$. If $\boldsymbol{v} \notin N_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}}$, it follows that its tangent component, $\boldsymbol{v}_\parallel = \operatorname{proj}_{T_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}}}(\boldsymbol{v})$, is non-zero.

However, since $\boldsymbol{v}_\perp := \boldsymbol{v} - \boldsymbol{v}_\parallel \in N_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}}$, we have

$$\begin{aligned}\mathbf{0} &= \mathbf{Im}\big\{\operatorname{diag}(A\boldsymbol{x})A^*\big\}\,\boldsymbol{v}_\perp \\ &= \mathbf{Im}\big\{\operatorname{diag}(A\boldsymbol{x})A^*\big\}\,\boldsymbol{v} - \mathbf{Im}\big\{\operatorname{diag}(A\boldsymbol{x})A^*\big\}\,\boldsymbol{v}_\parallel \\ &= -\mathbf{Im}\big\{\operatorname{diag}(A\boldsymbol{x})A^*\big\}\,\boldsymbol{v}_\parallel.\end{aligned}$$

Combine with the fact that $\boldsymbol{v}_\parallel \in T_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}}$, we deduce that

$$\begin{aligned}&\begin{cases}\mathbf{Re}\big\{\operatorname{diag}(A\boldsymbol{x})A^*\big\}\,\boldsymbol{v}_\parallel = \mathbf{0}, \\ \mathbf{Im}\big\{\operatorname{diag}(A\boldsymbol{x})A^*\big\}\,\boldsymbol{v}_\parallel = \mathbf{0}\end{cases} \\ &\Rightarrow \operatorname{diag}(A\boldsymbol{x})A^*\boldsymbol{v}_\parallel = \mathbf{0}.\end{aligned}$$

Since both $\operatorname{diag}(A\boldsymbol{x})$ and A^* are invertible (under Assumption 2), $\boldsymbol{v}_\parallel$ must be the zero vector. This is a contradiction. Therefore, $\boldsymbol{v}$ must be in $N_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}}$. The reverse inclusion is shown. Finally, the theorem is proven. ■

The normal-space characterization also admits a more direct proof, which we present below.

Proof. (Normal space characterization) Since for any real matrix M

$$\ker M = \Big(\operatorname{Col}(M^\mathsf{T})\Big)^\perp,$$

where $\operatorname{Col}(\cdot)$ returns the column space of a matrix and $(\cdot)^\perp$ returns the orthogonal complement of a subspace in $\mathbb{R}^{n\times 1}$, we obtain

$$\begin{aligned}&\ker\Big(\mathbf{Im}\big\{\operatorname{diag}(A\boldsymbol{x})A^*\big\}\Big) \\ &= \Big(\operatorname{Col}\Big(\mathbf{Im}\big\{A^*\operatorname{diag}(A\boldsymbol{x})\big\}\Big)\Big)^\perp \\ &= \left(\operatorname{span}\Big\{\mathbf{Im}\Big\{\boldsymbol{a}_k\mathrm{e}^{-\mathrm{i}t_k}\Big\}\Big\}_{k\in[n]}\right)^\perp \\ &= \Big(\operatorname{span}\left\{-\sin t_k\boldsymbol{u}_k + \cos t_k\boldsymbol{v}_k\right\}_{k\in\mathcal{N}_c}\Big)^\perp.\end{aligned}$$

This is equivalent to stating

$$\ker\Big(\mathbf{Im}\big\{\operatorname{diag}(A\boldsymbol{x})A^*\big\}\Big) = \left(\operatorname{span}\left\{\frac{\partial\boldsymbol{x}}{\partial t_k}\right\}_{k\in\mathcal{N}_c}\right)^\perp.$$

Notice that the term in $(\cdot)^\perp$ is exactly the tangent space. By the definition of the normal space, viz.,

$$N_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}} := \left(T_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}}\right)^\perp,$$

we obtain

$$N_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}} = \ker\Big(\mathbf{Im}\big\{\operatorname{diag}(A\boldsymbol{x})A^*\big\}\Big).$$

The result is again obtained. ■

From the proofs above, it is a simple exercise to derive several equivalent characterizations of the tangent and normal spaces:

$$\begin{aligned}T_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}} &= \ker\Big(\mathbf{Re}\big\{\operatorname{diag}(A\boldsymbol{x})A^*\big\}\Big) &\text{(A4)}\\ &= \operatorname{Col}\Big(\mathbf{Im}\big\{A\operatorname{diag}(A^*\boldsymbol{x})\big\}\Big) &\text{(A5)}\\ &= \operatorname{span}\left\{\frac{\partial\boldsymbol{x}}{\partial t_k}\right\}_{k\in\mathcal{N}_c} &\text{(A6)}\end{aligned}$$

and

$$\begin{aligned}N_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}} &= \ker\Big(\mathbf{Im}\big\{\operatorname{diag}(A\boldsymbol{x})A^*\big\}\Big) &\text{(A7)}\\ &= \operatorname{Col}\Big(\mathbf{Re}\big\{A\operatorname{diag}(A^*\boldsymbol{x})\big\}\Big) &\text{(A8)}\\ &= \operatorname{span}\left\{\frac{\partial\boldsymbol{x}}{\partial y_k}\right\}_{k\in\mathcal{N}_p\cup\mathcal{N}_c}. &\text{(A9)}\end{aligned}$$

It should also be noted that the characterizations we provide here are equivalent to those given by Barnett et al. [1]:

$$\begin{aligned}T_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}} &= \operatorname{span}\left\{(P - P^{-1})\boldsymbol{x}\;\middle|\; P \text{ is a translation}\right\}, \\ N_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}} &= \operatorname{span}\left\{(P + P^{-1})\boldsymbol{x}\;\middle|\; P \text{ is a translation}\right\}.\end{aligned}$$

The equivalence can be readily shown by observing that the eigenvectors of P are the $\boldsymbol{a}_k$'s. However, we will not delve into the details.

Theorem 5. *For all $(\boldsymbol{x}, \boldsymbol{v}) \in T\mathbb{A}_{\boldsymbol{y}} \oplus N\mathbb{A}_{\boldsymbol{y}}$, the pushforward map $(P_\mathcal{A})_{*,\boldsymbol{x}}$ is given by*

$$(P_\mathcal{A})_{*,\boldsymbol{x}}(\boldsymbol{v}) := \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} P_\mathcal{A}(\boldsymbol{x} + t\boldsymbol{v}) = \operatorname{proj}_{T_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}}}(\boldsymbol{v}). \quad (16)$$

Proof. We directly calculate the differentiation above using Eqn. (11):

$$\begin{aligned}\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_0 P_\mathcal{A}(\boldsymbol{x} + t\boldsymbol{v}) &= \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_0 A^\mathsf{H}\left(\boldsymbol{y}\odot\operatorname{sgn}(A(\boldsymbol{x}+t\boldsymbol{v}))\right) \\ &= A^\mathsf{H}\left(\boldsymbol{y}\odot\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_0 \operatorname{sgn}(A(\boldsymbol{x}+t\boldsymbol{v}))\right).\end{aligned}$$

Note that since[1]

$$\begin{aligned}\frac{\partial\operatorname{sgn}(\boldsymbol{z})}{\partial\boldsymbol{z}}[\boldsymbol{w}] &= \frac{\partial}{\partial\boldsymbol{z}}\left(\frac{\boldsymbol{z}}{|\boldsymbol{z}|}\right)[\boldsymbol{w}] \\ &= \frac{\boldsymbol{w}}{|\boldsymbol{z}|} - \frac{\boldsymbol{z}}{2|\boldsymbol{z}|^3}\odot(\boldsymbol{z}^*\odot\boldsymbol{w} + \boldsymbol{z}\odot\boldsymbol{w}^*) \\ &= \frac{\boldsymbol{w}}{2|\boldsymbol{z}|} - \frac{\boldsymbol{z}^2\odot\boldsymbol{w}^*}{2|\boldsymbol{z}|^3},\end{aligned}$$

[1]The complex directional derivatives introduced here are Gâteaux/Fréchet derivatives, not to be confused with the Wirtinger derivatives.

we have

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_0 \operatorname{sgn}(A(\boldsymbol{x}+t\boldsymbol{v})) = \left.\frac{\partial \operatorname{sgn}(\boldsymbol{z})}{\partial \boldsymbol{z}}\right|_{\boldsymbol{z}=A\boldsymbol{x}} [A\boldsymbol{v}]$$
$$= \frac{A\boldsymbol{v}}{2|A\boldsymbol{x}|} - \frac{(A\boldsymbol{x})^2 \odot (A^*\boldsymbol{v})}{2|A\boldsymbol{x}|^3}$$
$$= \frac{\mathrm{i}A\boldsymbol{x}}{|A\boldsymbol{x}|^3} \odot \mathbf{Im}\{(A^*\boldsymbol{x}) \odot (A\boldsymbol{v})\}.$$

Next, the vector $\boldsymbol{v}$ can be orthogonally decomposed into $\boldsymbol{v} = \boldsymbol{v}_\parallel + \boldsymbol{v}_\perp$, where $\boldsymbol{v}_\parallel \in T_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}}$ and $\boldsymbol{v}_\perp \in N_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}}$. Then, by Theorem 4,

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_0 \operatorname{sgn}(A(\boldsymbol{x}+t\boldsymbol{v})) = \frac{\mathrm{i}A\boldsymbol{x}}{|A\boldsymbol{x}|^3} \odot \mathbf{Im}\Big\{(A^*\boldsymbol{x}) \odot (A\boldsymbol{v}_\parallel)\Big\}.$$

This can be further simplified by noting that

$$\mathbf{Re}\Big\{(A^*\boldsymbol{x}) \odot (A\boldsymbol{v}_\parallel)\Big\} = \mathbf{0}$$
$$\Leftrightarrow (A^*\boldsymbol{x}) \odot (A\boldsymbol{v}_\parallel) = -(A\boldsymbol{x}) \odot (A^*\boldsymbol{v}_\parallel)$$
$$\Leftrightarrow \mathrm{i} \cdot \mathbf{Im}\Big\{(A^*\boldsymbol{x}) \odot (A\boldsymbol{v}_\parallel)\Big\} = (A^*\boldsymbol{x}) \odot (A\boldsymbol{v}_\parallel),$$

hence,

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_0 \operatorname{sgn}(A(\boldsymbol{x}+t\boldsymbol{v})) = \frac{A\boldsymbol{x}}{|A\boldsymbol{x}|^3} \odot (A^*\boldsymbol{x}) \odot (A\boldsymbol{v}_\parallel) = \frac{A\boldsymbol{v}_\parallel}{|A\boldsymbol{x}|}.$$

Plug this back into the pushforward map, we obtain

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_0 P_{\mathcal{A}}(\boldsymbol{x}+t\boldsymbol{v}) = A^{\mathsf{H}}\left(\boldsymbol{y} \odot \frac{A\boldsymbol{v}_\parallel}{|A\boldsymbol{x}|}\right)$$
$$= A^{\mathsf{H}}(A\boldsymbol{v}_\parallel)$$
$$= \boldsymbol{v}_\parallel = \operatorname{proj}_{T_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}}}(\boldsymbol{v}).$$

Thus, it is shown. ∎

This result actually holds more generally [2, Lemma 3.1]: for any nearest-point projection onto a manifold, the pushforward of the projection is the projection onto the tangent space. Nevertheless, we provide a first-principles proof of this special case because the derivation is short and transparent.

Corollary 4. *$P_{\mathcal{A}} : T\mathbb{A}_{\boldsymbol{y}} \to \mathbb{A}_{\boldsymbol{y}}$ given by $(\boldsymbol{x}, \boldsymbol{v}) \mapsto P_{\mathcal{A}}(\boldsymbol{x}+\boldsymbol{v})$ is a retraction.*

Proof. A retraction R on a manifold $\mathcal{M}$, by definition (see Definition 4.1.1 in [3]), satisfies (a) $R(x, \mathbf{0}) = x$ and (b) $R_{*,x}(\cdot) = \mathrm{id}_{T_x\mathcal{M}}$ for all $x \in \mathcal{M}$.

The property (a) is automatically satisfied by $P_{\mathcal{A}}$ since it is a projection. The property (b) is again satisfied by utilizing Theorem 5 with $\boldsymbol{v} \in T_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}}$ to show $\operatorname{proj}_{T_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}}}(\boldsymbol{v}) = \boldsymbol{v}$. ∎

Corollary 5. *For sufficiently small μ, the stationarity conditions of `TorusGD` and `dualPGD` coincide.*

Proof. For μ small enough such that no wrap-around happens, the stationarity condition of `TorusGD` is $\nabla(f \circ \boldsymbol{x})(\boldsymbol{t}) = \mathbf{0}$, while the stationarity condition of `dualPGD` is $\nabla F(\boldsymbol{x}) \in N_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}}$.

Since for any vector $\boldsymbol{v}$ of consistent size,

$$\frac{\partial}{\partial \boldsymbol{t}}(f \circ \boldsymbol{x})[\boldsymbol{v}] = \frac{\partial F}{\partial \boldsymbol{x}} \circ \frac{\partial \boldsymbol{x}}{\partial \boldsymbol{t}}[\boldsymbol{v}],$$

we have

$$\nabla(f \circ \boldsymbol{x})(\boldsymbol{t}) = S^{\mathsf{T}}\nabla F(\boldsymbol{x}),$$

where $\boldsymbol{x} = \boldsymbol{x}(\boldsymbol{t})$ and $S := \partial\boldsymbol{x}/\partial\boldsymbol{t} \in \mathbb{R}^{n\times d}$. Therefore,

$$\nabla(f \circ \boldsymbol{x})(\boldsymbol{t}) = \mathbf{0} \Leftrightarrow \nabla F(\boldsymbol{x}) \in \ker(S^{\mathsf{T}}).$$

Further note that

$$\ker(S^{\mathsf{T}}) = \big(\mathrm{Col}(S)\big)^\perp = \big(T_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}}\big)^\perp = N_{\boldsymbol{x}}\mathbb{A}_{\boldsymbol{y}}.$$

Thus, it is shown. ∎

As a final remark, let us derive Eqn. (15), the Hessian of $f \circ \boldsymbol{x}$.

$$\nabla^2(f \circ \boldsymbol{x}) = S^{\mathsf{T}}\nabla^2 F(\boldsymbol{x})S + \operatorname{diag}\Big(D^{\mathsf{T}}\nabla F(\boldsymbol{x})\Big) \tag{15}$$

Proof. Since

$$\nabla(f \circ \boldsymbol{x})(\boldsymbol{t}) = S^{\mathsf{T}}\nabla F(\boldsymbol{x}),$$

we have, for any vector $\boldsymbol{v}$ of consistent size,

$$\boldsymbol{v}^{\mathsf{T}}\nabla^2(f \circ \boldsymbol{x})\boldsymbol{v} = \frac{\partial}{\partial \boldsymbol{t}}\Big((S\boldsymbol{v})^{\mathsf{T}}\nabla F(\boldsymbol{x})\Big)[\boldsymbol{v}]$$
$$= \underbrace{\boldsymbol{v}^{\mathsf{T}}\frac{\partial S^{\mathsf{T}}}{\partial \boldsymbol{t}}[\boldsymbol{v}]\nabla F(\boldsymbol{x})}_{(\star)} + \underbrace{(S\boldsymbol{v})^{\mathsf{T}}\frac{\partial}{\partial \boldsymbol{t}}\big(\nabla F(\boldsymbol{x})\big)[\boldsymbol{v}]}_{(*)}.$$

Define $D := \mathrm{row}\{\partial^2\boldsymbol{x}/\partial t_k^2\}_{k\in\mathcal{N}_c}$, the two terms are

$$(\star) = \boldsymbol{v}^{\mathsf{T}}\left(\mathrm{row}\left\{\frac{\partial}{\partial \boldsymbol{t}}\frac{\partial \boldsymbol{x}}{\partial t_k}[\boldsymbol{v}]\right\}_{k\in\mathcal{N}_c}\right)^{\mathsf{T}}\nabla F(\boldsymbol{x})$$
$$= \boldsymbol{v}^{\mathsf{T}}\left(\mathrm{row}\left\{\frac{\partial^2 \boldsymbol{x}}{\partial t_k^2}[v_k]\right\}_{k\in\mathcal{N}_c}\right)^{\mathsf{T}}\nabla F(\boldsymbol{x})$$
$$= \boldsymbol{v}^{\mathsf{T}}\big(D\operatorname{diag}(\boldsymbol{v})\big)^{\mathsf{T}}\nabla F(\boldsymbol{x})$$
$$= \boldsymbol{v}^{\mathsf{T}}\operatorname{diag}(\boldsymbol{v})D^{\mathsf{T}}\nabla F(\boldsymbol{x})$$
$$= \boldsymbol{v}^{\mathsf{T}}\operatorname{diag}\Big(D^{\mathsf{T}}\nabla F(\boldsymbol{x})\Big)\boldsymbol{v}$$

and

$$(*) = (S\boldsymbol{v})^{\mathsf{T}}\frac{\partial}{\partial \boldsymbol{x}}\big(\nabla F(\boldsymbol{x})\big) \circ \frac{\partial \boldsymbol{x}}{\partial \boldsymbol{t}}[\boldsymbol{v}]$$
$$= (S\boldsymbol{v})^{\mathsf{T}}\nabla^2 F(\boldsymbol{x})(S\boldsymbol{v}).$$

Combining the two equations, we obtain Eqn. (15). ∎

Table B1: Retrieval rate of noisy signals over 1000 trials.

Noise Level (r)	Algorithm	Retrieval Rate (%)		
		3(b)	3(c)	3(d)
0.01	dualPGD	98.9	6.6	85.8
	CHIO	98.5	71.0	100.0
0.04	dualPGD	98.6	4.9	83.8
	CHIO	98.5	71.9	100.0
0.0625	dualPGD	98.2	3.1	77.4
	CHIO	98.7	76.1	100.0

B. FURTHER NUMERICAL SIMULATIONS

In Sec. 5 of the manuscript, we considered noiseless binary signals in our preliminary experiments to demonstrate the effectiveness of our algorithms. However, the performance on certain test signals, most notably the one shown in Fig. 3(c) of the manuscript, indicates that the iterates may suffer from stagnation at local minima. In this section, we address the issue of stagnation at local minima and further evaluate the robustness of our algorithms under noisy settings. The source code for the simulations can be found via the link `https://github.com/WenPerng/Manifold-Optimization-FPR`.

B.1. Escaping Local Minima

In this subsection, we tested a variant of `dualPGD` that combined momentum with random restarts.

Specifically, we incorporated momentum with a decay factor $\beta = 0.9$ and implemented a random restart strategy that was triggered whenever the iteration process stagnated. To maintain computational efficiency, the total number of restarts per trial was limited to 15.

We evaluated this approach on the 61×61 "ICIP" logo (test signal Fig. 3(c) of the manuscript), and observed a significant improvement in the retrieval rate, from 8.0% to 37.0%.

This result highlights the framework's high compatibility with modern optimization techniques. Since the proposed vanilla `dualPGD` is essentially a Riemannian gradient descent algorithm, we can incorporate techniques such as momentum methods, stochastic noise, or learned stepsize scheduling into `dualPGD` to further enhance its performance.

B.2. Noisy Signals

In many applications of Fourier phase retrieval, the observed Fourier magnitude is often corrupted with noise. Here we tested the applicability of `dualPGD` to retrieve a binary signal under the noise model of

$$\boldsymbol{y} = |A\boldsymbol{x} + \boldsymbol{z}| \,,$$

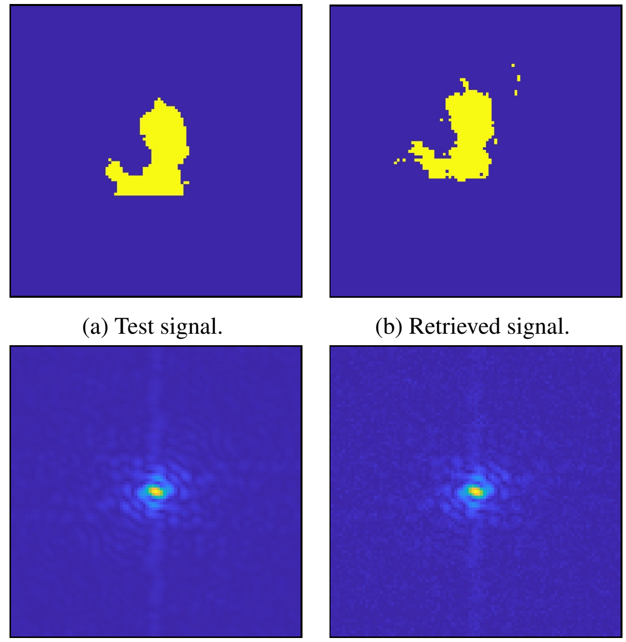
where $\boldsymbol{z}$ is a realization of a zero-mean Gaussian random vector.

We tested our simulation over different noise levels, characterized via the ratio defined as

$$r := \frac{\|\boldsymbol{z}\|_2^2}{\|\boldsymbol{x}\|_2^2}.$$

The experimental results in Table B1 demonstrate that the proposed `dualPGD` is reasonably robust in the presence of noise. Under a noise level of $r = 0.01$, `dualPGD` maintains a retrieval rate close to the noiseless case, achieving 98.9%, 6.6%, and 85.8% for the respective test signals (which are Fig. 3(b) to 3(d) of the manuscript). Although the success rate naturally declines as the noise intensity increases (e.g., to $r = 0.0625$), the performance remains stable across different configurations. See Fig. B1 for an instance of a successful retrieval with noisy observed magnitude.

These preliminary results indicate that our algorithm can handle certain corrupted Fourier magnitude measurements without noise-specific tuning. For future work, the robustness of the framework could be further analyzed.



(a) Test signal. (b) Retrieved signal. (c) Noiseless Fourier magnitude. (d) Noisy Fourier magnitude.

Fig. B1: Fourier phase retrieval of a test signal with noisy observed Fourier magnitude. (b) is an instance of a retrieved binary signal via `dualPGD` using the magnitude in (d). The noise level of (d) is $r = 0.0625$.

C. FINAL REMARKS

The first remark elaborates on Eqn. (A3) and clarifies the connection between the properties of $\ell(\boldsymbol{x})$ and the Polyak-Łojasiewicz condition. The second remark concerns how we determine the initialization of our algorithms. The initialization actually affects the optimization domain of our algorithms. The last remark addresses how the Hessian is computed to ensure a fast code execution.

C.1. ℓ is Polyak-Łojasiewicz

The function $\ell(\boldsymbol{x}) = \big\| |A\boldsymbol{x}| - \boldsymbol{y} \big\|_2^2$ satisfies

$$\frac{1}{2}\big\|\nabla\ell(\boldsymbol{x})\big\|^2 = 2\ell(\boldsymbol{x}). \tag{C1}$$

This fact immediately leads to the second statement of Theorem 1:

$$\begin{aligned}
&\because\ \frac{1}{2}\big\|\nabla\ell(\boldsymbol{x})\big\|^2 = 2\ell(\boldsymbol{x}),\\
&\therefore\ \nabla\left(\frac{1}{2}\big\|\nabla\ell(\boldsymbol{x})\big\|^2\right) = H(\boldsymbol{x})\cdot\nabla\ell(\boldsymbol{x}) = 2\nabla\ell(\boldsymbol{x}).
\end{aligned}$$

Furthermore, ℓ is also 2-PŁ [4] and the properties for PŁ functions are automatically satisfied by ℓ, although the equality above is much stronger than the PŁ inequality.

One such property is that PŁ functions are bounded below by square distance functions [5]. For the case of ℓ, it is exactly a square distance function to a set, namely $\mathbb{A}_{\boldsymbol{y}}$.

Proof. By Corollary 3,

$$\begin{aligned}
\operatorname{dist}(\boldsymbol{x}, \mathbb{A}_{\boldsymbol{y}})^2 &= \min_{\boldsymbol{x}'\in\mathbb{A}_{\boldsymbol{y}}} \big\|\boldsymbol{x}-\boldsymbol{x}'\big\|_2^2\\
&= \big\|\boldsymbol{x} - P_{\mathcal{A}}(\boldsymbol{x})\big\|_2^2\\
&= \Big\|\boldsymbol{x} - A^{-1}\left(\operatorname{sgn}(A\boldsymbol{x})\odot\boldsymbol{y}\right)\Big\|_2^2\\
&= \big\|A\boldsymbol{x} - \operatorname{sgn}(A\boldsymbol{x})\odot\boldsymbol{y}\big\|_2^2\\
&= \big\|(A\boldsymbol{x})\odot\operatorname{sgn}(A^*\boldsymbol{x}) - \boldsymbol{y}\big\|_2^2\\
&= \big\||A\boldsymbol{x}| - \boldsymbol{y}\big\|_2^2\\
&= \ell(\boldsymbol{x}).
\end{aligned}$$

Thus, it is shown. ∎

C.2. On the Optimization Domain

For both `TorusGD` and `dualPGD`, all iterates remain on the same torus as the initialization $\boldsymbol{x}_0$. Let T be a connected component of $\mathbb{A}_{\boldsymbol{y}}$ with $\boldsymbol{x}_0 \in T$, the two algorithms are essentially solving for the problem

$$\min_{\boldsymbol{x}\in T\subset\mathbb{A}_{\boldsymbol{y}}} f(\boldsymbol{x}), \tag{C2}$$

where f measures the mismatch between $\boldsymbol{x}$ and the signal-domain constraint set $\mathcal{B}$. How do we determine which torus T is?

For our numerical simulations in Sec. 5 of the manuscript, $\mathcal{B}$ is the set of binary signals. One can show that, in order for T to contain a global minimizer of f, it is necessary that T contain at least one of the trivial associates resulting from translations and reflection (all but the global sign-flip trivial associates). These are the trivial associates that we want to retrieve since they globally minimize f.

Interestingly, the sizes of test signals we have chosen are all of odd lengths. Hence,

$$\big|\mathcal{N}_p\big| = 2^0 = 1, \tag{C3}$$

and there are a total of $2^{|\mathcal{N}_p|} = 2$ tori. One torus contains all the trivial associates resulting from translations and reflection, and the other contains the global sign-flip ones.

In the code implementation, we simply sample $\boldsymbol{x}_0$ from

$$\boldsymbol{x}_0 \sim P_{\mathcal{A}}\left(\operatorname{Unif}\left([0,1]^{n\times 1}\right)\right). \tag{C4}$$

By Corollary 3 and the fact that the two tori have a separating hyperplane $\mathbf{1}^\top\boldsymbol{x} = 0$, it is guaranteed that the $\boldsymbol{x}_0$ resulting from Eqn. (C4) lies on the torus that contains all trivial associates resulting from translations and reflection.

C.3. Computation of the Hessian

The computation of the Hessian $\nabla^2(f\circ\boldsymbol{x})$ (Eqn. (15)) requires the computation of the matrices

$$S := \frac{\partial\boldsymbol{x}}{\partial\boldsymbol{t}} = \operatorname{row}\left\{\frac{\partial\boldsymbol{x}}{\partial t_k}\right\}_{k\in\mathcal{N}_c} \in \mathbb{R}^{n\times d}$$

and

$$D := \operatorname{row}\left\{\frac{\partial^2\boldsymbol{x}}{\partial t_k^2}\right\}_{k\in\mathcal{N}_c} \in \mathbb{R}^{n\times d},$$

where $\operatorname{row}\{\cdot\}$ is the row-wise stacking operator.

Due to their large sizes, we avoid explicit matrix multiplication when computing these two matrices. Instead, one can utilize the basis $\{\boldsymbol{u}_k, \boldsymbol{v}_k\}_{k\in\mathcal{N}_p\cup\mathcal{N}_c}$ and orthogonally decompose $\boldsymbol{x}$ to obtain $\boldsymbol{s}$ and $\boldsymbol{t}$. The computation of S and D is then fast and direct using $\boldsymbol{y}$, $\boldsymbol{s}$, $\boldsymbol{t}$, $\{\boldsymbol{u}_k, \boldsymbol{v}_k\}_{k\in\mathcal{N}_p\cup\mathcal{N}_c}$, and the parameterization established in Theorem 3.

D. REFERENCES

[1] A. Barnett, C. L. Epstein, L. Greengard, and J. Magland, "Geometry of the phase retrieval problem," 2020.

[2] P.-A. Absil and Jérôme Malick, "Projection-like retractions on matrix manifolds," *SIAM Journal on Optimization*, vol. 22, no. 1, pp. 135–158, 2012.

[3] P.-A. Absil, R. Mahony, and R. Sepulchre, *Optimization Algorithms on Matrix Manifolds*, Princeton University Press, Princeton, NJ, 2008.

[4] B.T. Polyak, "Gradient methods for the minimisation of functionals," *USSR Computational Mathematics and Mathematical Physics*, vol. 3, no. 4, pp. 864–878, 1963.

[5] Guillaume Garrigos, "Square distance functions are polyak-łojasiewicz and vice-versa," 2023.